

InverseNet: Benchmarking Operator Mismatch and Calibration Across Compressive Imaging Modalities

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Abstract. Compressive imaging systems rely on accurate knowledge of the forward measurement operator for high-quality reconstruction. In practice, manufacturing tolerances, assembly errors, and environmental drift introduce mismatch between the assumed and true operators, degrading reconstruction quality by 10–20 dB. Despite this, existing benchmarks evaluate reconstruction methods exclusively under ideal operator assumptions. We present **InverseNet**, a unified benchmark for quantifying operator mismatch degradation and oracle calibration recovery across three compressive imaging modalities: coded aperture snapshot spectral imaging (CASSI), coded aperture compressive temporal imaging (CACTI), and single-pixel camera (SPC). Our benchmark evaluates 11 reconstruction methods—spanning classical optimization and deep learning—under a standardized three-scenario protocol: (I) ideal operator, (II) mismatched operator, and (III) oracle-corrected operator. Across 27 test scenes and 11 methods, we find that operator mismatch degrades deep learning methods by 10–21 dB, with mask-aware architectures both the most sensitive and the most recoverable through calibration (up to 75% recovery for CASSI, 90% for SPC). Classical methods show greater robustness but lower peak quality. We release all reconstruction arrays, metrics, and figure-generation code to support reproducible research in operator-aware computational imaging.

Keywords: Compressive imaging · Operator mismatch · Calibration · Benchmark · Spectral imaging · Video compressive sensing · Single-pixel camera

1 Introduction

Compressive imaging acquires fewer measurements than the Nyquist limit by exploiting signal structure, recovering the full signal through computational reconstruction. This paradigm underlies diverse modalities including hyperspectral imaging via coded apertures [1,2], video compressive sensing via temporal coding [3,4], and single-pixel cameras via structured illumination [5,6]. In all cases, reconstruction quality depends critically on knowledge of the forward measurement operator—the mapping from scene to measurements.

In practice, the assumed forward operator inevitably differs from the true physical operator. For CASSI systems, mask misalignment of even 0.5 pixels

and 0.1° rotation between the coded aperture and the detector can degrade peak signal-to-noise ratio (PSNR) by over 10 dB [1]. For CACTI, clock drift, duty cycle variations, and detector gain offsets collectively introduce multi-parameter mismatch. For single-pixel cameras, gain drift in the digital micromirror device or photodetector causes systematic measurement errors. These *operator mismatches* are ubiquitous in deployed systems yet are systematically ignored in reconstruction benchmarks.

The benchmark gap. Existing benchmarks for compressive imaging reconstruction—such as the KAIST hyperspectral dataset for CASSI [17] and the SCI video benchmark for CACTI [4]—assume perfect operator knowledge. Methods are evaluated only under Scenario I (ideal operator), providing no information about robustness to operator mismatch or the potential benefit of calibration. This creates a critical blind spot: a method that achieves state-of-the-art PSNR under ideal conditions may catastrophically fail in practice if it is highly sensitive to operator errors.

Contributions. We address this gap with InverseNet, which makes three contributions:

1. **Unified three-scenario protocol.** We define a standardized evaluation framework with three scenarios—ideal (I), mismatched (II), and oracle-corrected (III)—applicable across compressive imaging modalities. The gap between Scenarios I and II quantifies mismatch sensitivity; the recovery from II to III quantifies the upper bound of calibration benefit.
2. **Cross-modality benchmark.** We evaluate 11 reconstruction methods (4 CASSI, 4 CACTI, 3 SPC) spanning classical optimization (GAP-TV, FISTA-TV) and deep learning (MST, HDNet, EfficientSCI, ELP-Unfolding, ISTA-Net, HATNet) across 27 test scenes, producing over 240 reconstruction experiments.
3. **Open dataset.** We release all reconstruction arrays, per-scene metrics (PSNR, SSIM, SAM), and analysis code to enable reproducible operator-mismatch research.

Our key findings include: (a) operator mismatch degrades deep learning methods by 10–21 dB while classical methods lose only 0.1–11 dB; (b) mask-aware architectures (MST, ELP-Unfolding) are simultaneously the most sensitive to mismatch and the most recoverable through calibration; (c) mask-oblivious architectures (HDNet) show zero calibration benefit; (d) CACTI exhibits the most severe mismatch degradation (up to 20.58 dB) due to its multi-parameter mismatch space.

2 Related Work

Compressive imaging reconstruction. Classical reconstruction methods for compressive imaging employ convex optimization with sparsity-promoting regularization. GAP-TV [7] uses the generalized alternating projection framework with total variation (TV) regularization, applicable to both CASSI and CACTI.

FISTA [8] and ADMM [9] provide efficient solvers for ℓ_1 -regularized inverse problems. Deep learning methods have dramatically improved reconstruction quality: MST [10] introduces mask-guided spectral transformers for CASSI; HDNet [11] uses dual-domain processing; EfficientSCI [12] and ELP-Unfolding [13] target video compressive sensing; ISTA-Net [14] and HATNet [15] address single-pixel imaging. All these methods are developed and evaluated assuming perfect forward operators.

Calibration and operator mismatch. Operator mismatch has been studied in specific modalities but not systematically benchmarked. For CASSI, Wagadarikar et al. [1] identified mask misalignment as a key error source, and subsequent work [21] proposed calibration procedures. For MRI, the fastMRI benchmark [19] evaluates undersampling patterns but assumes known coil sensitivities. Phase retrieval literature has examined model mismatch in coherent imaging [20]. However, no prior work provides a unified cross-modality benchmark quantifying both the degradation from mismatch and the recovery potential from calibration.

Reconstruction benchmarks. The KAIST TSA dataset [17] provides 10 simulated hyperspectral scenes for CASSI evaluation. The SCI video benchmark [4] provides grayscale video sequences for CACTI. The Set11 dataset [18] is standard for single-pixel camera evaluation. These benchmarks evaluate reconstruction quality under ideal conditions only. InverseNet extends them by introducing controlled operator mismatch and measuring calibration recovery, creating the first cross-modality operator-mismatch benchmark.

3 The InverseNet Benchmark

3.1 Unified Three-Scenario Protocol

We define three evaluation scenarios that apply uniformly across all compressive imaging modalities. Let Φ denote the true (physical) forward operator and $\hat{\Phi}$ the assumed (nominal) operator used during reconstruction.

- **Scenario I (Ideal):** The measurement is formed with the ideal operator $\hat{\Phi}$, and reconstruction uses the same ideal operator. This represents the best-case performance with perfect operator knowledge: $\mathbf{y} = \hat{\Phi}\mathbf{x} + \mathbf{n}$, reconstruct with $\hat{\Phi}$.
- **Scenario II (Baseline):** The measurement is formed with the true (mismatched) operator Φ , but reconstruction still uses the assumed operator $\hat{\Phi}$. This represents the realistic deployment scenario where the physical operator has drifted from its assumed value: $\mathbf{y} = \Phi\mathbf{x} + \mathbf{n}$, reconstruct with $\hat{\Phi}$.
- **Scenario III (Oracle):** The measurement is formed with the true operator Φ (same as Scenario II), but reconstruction uses the true operator as oracle knowledge: $\mathbf{y} = \Phi\mathbf{x} + \mathbf{n}$, reconstruct with Φ . This represents the upper bound achievable through perfect calibration.

This protocol yields two diagnostic metrics per method:

$$\Delta_{\text{deg}} = \text{PSNR}_{\text{I}} - \text{PSNR}_{\text{II}} \quad (\text{mismatch degradation}), \quad (1)$$

$$\Delta_{\text{rec}} = \text{PSNR}_{\text{III}} - \text{PSNR}_{\text{II}} \quad (\text{oracle recovery}), \quad (2)$$

and the *recovery ratio* $\rho = \Delta_{\text{rec}}/\Delta_{\text{deg}} \in [0, 1]$, which measures what fraction of the mismatch loss can be recovered through calibration.

3.2 CASSI: Coded Aperture Snapshot Spectral Imaging

Forward model. CASSI acquires a 2D measurement $\mathbf{y} \in \mathbb{R}^{H \times W'}$ of a 3D hyperspectral cube $\mathbf{x} \in \mathbb{R}^{H \times W \times \Lambda}$ through a coded aperture mask $\mathbf{M} \in \{0, 1\}^{H \times W}$ followed by a dispersive prism. The measurement at pixel (i, j) is:

$$y(i, j) = \sum_{\lambda=1}^{\Lambda} M(i, j - d(\lambda)) \cdot x(i, j, \lambda) + n(i, j), \quad (3)$$

where $d(\lambda)$ is the dispersion shift for spectral band λ and $W' = W + (\Lambda - 1) \cdot s$ with dispersion step s .

Mismatch model. We model CASSI operator mismatch as a 3-parameter spatial perturbation of the coded aperture mask:

$$\Phi = \mathcal{T}(dx, dy, \theta) \circ \hat{\Phi}, \quad (4)$$

where dx, dy are subpixel translational shifts and θ is a rotational misalignment. We use $dx = 0.5$ px, $dy = 0.3$ px, $\theta = 0.1^\circ$, representing moderate assembly tolerances.

Reconstruction methods. We evaluate four methods: **GAP-TV** [7]: classical accelerated proximal gradient with TV regularization (100 iterations, $\lambda_{\text{TV}} = 4.0$); **HDNet** [11]: dual-domain deep network with spectral discrimination learning (pretrained); **MST-S** [10]: mask-guided spectral transformer, small variant (2 stages, blocks [2, 2, 2], pretrained); **MST-L** [10]: mask-guided spectral transformer, large variant (2 stages, blocks [4, 7, 5], pretrained).

Dataset. We use 10 scenes from the KAIST TSA simulated dataset [17], each consisting of a $256 \times 256 \times 28$ hyperspectral cube spanning 450–650 nm. Measurements are formed with a binary random mask ($s = 2$ pixels/band), yielding 256×310 detector images. Low noise ($\alpha = 10^5$ photon peak, $\sigma = 0.01$ read noise) isolates the effect of operator mismatch.

3.3 CACTI: Coded Aperture Compressive Temporal Imaging

Forward model. CACTI acquires a single 2D snapshot $\mathbf{y} \in \mathbb{R}^{H \times W}$ encoding B high-speed video frames $\mathbf{x} \in \mathbb{R}^{H \times W \times B}$ through a dynamic coded aperture:

$$y(i, j) = \sum_{b=1}^B C_b(i, j) \cdot x(i, j, b) + n(i, j), \quad (5)$$

where $C_b \in \{0, 1\}^{H \times W}$ is the binary mask pattern for temporal frame b .

Mismatch model. CACTI mismatch involves 8 parameters capturing spatial, temporal, and radiometric errors: spatial shifts ($dx = 0.5$ px, $dy = 0.3$ px), rotation ($\theta = 0.1^\circ$), temporal clock offset ($\Delta t = 0.05$), duty cycle deviation ($\eta = 0.95$), detector gain ($g = 1.02$), offset ($o = 0.002$), and measurement noise ($\sigma_n = 1.0$).

Reconstruction methods. We evaluate four methods: **GAP-TV** [7]: classical iterative with TV regularization; **PnP-FFDNet** [16]: plug-and-play with FFDNet denoiser; **ELP-Unfolding** [13]: efficient long-range perception unfolding network (pretrained); **EfficientSCI** [12]: efficient deep learning for SCI (pretrained).

Dataset. We use 6 standard benchmark videos (*kobe*, *traffic*, *runner*, *drop*, *crash*, *aerial*) at 256×256 resolution with $B = 8$ temporal frames per snapshot, following the standard SCI evaluation protocol [4].

3.4 SPC: Single-Pixel Camera

Forward model. The single-pixel camera acquires m scalar measurements of an image $\mathbf{x} \in \mathbb{R}^n$ through structured illumination patterns:

$$\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{n}, \quad (6)$$

where $\mathbf{A} \in \mathbb{R}^{m \times n}$ is the measurement matrix (typically Gaussian or Hadamard patterns) with compression ratio m/n .

Mismatch model. We model SPC mismatch as multiplicative gain drift affecting the measurement matrix:

$$\Phi = \text{diag}(1 + \alpha \cdot \mathbf{g}) \cdot \hat{\Phi}, \quad (7)$$

where $\alpha = 0.0015$ controls the drift magnitude and $\mathbf{g}$ is a per-row gain perturbation vector. Additional measurement noise $\sigma_y = 0.03$ is applied.

Reconstruction methods. We evaluate three methods: **FISTA-TV** [8]: fast iterative shrinkage-thresholding with TV regularization (500 iterations, $\lambda = 0.005$); **ISTA-Net** [14]: learned iterative shrinkage-thresholding network (pretrained); **HATNet** [15]: hybrid attention transformer for compressed sensing (pretrained).

Dataset. We use the 11 standard Set11 test images (*Monarch*, *Parrots*, *barbara*, *boats*, *cameraman*, *fingerprint*, *flinstones*, *foreman*, *house*, *lena256*, *peppers256*) at 256×256 resolution with 25% sampling ratio.

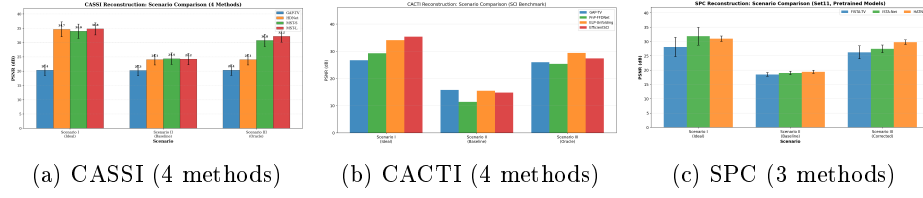


Fig. 1: PSNR across three scenarios for all modalities. Scenario I (Ideal): perfect operator. Scenario II (Baseline): mismatched operator. Scenario III (Oracle): true operator used for reconstruction. The collapse of deep learning methods under Scenario II is visible across all modalities, with CACTI showing the most severe degradation.

Table 1: CASSI reconstruction results (10 KAIST scenes, $256 \times 256 \times 28$). PSNR (dB) / SSIM reported as mean $\pm$ std. Δ_{deg} : degradation (I $\rightarrow$ II). Δ_{rec} : oracle recovery (II $\rightarrow$ III). ρ : recovery ratio. Best recovery in **green**. † Recovery ratio undefined due to near-zero degradation.

Method	Scenario I	Scenario II	Scenario III	Δ_{deg}	Δ_{rec}	ρ
GAP-TV [7]	20.37 $\pm$ 1.84 / .620	20.28 $\pm$ 1.83 / .602	20.38 $\pm$ 1.84 / .620	0.08	0.09	N/A †
HDNet [11]	34.66 $\pm$ 2.62 / .970	24.05 $\pm$ 1.85 / .813	24.05 $\pm$ 1.85 / .813	10.61	0.00	0%
MST-S [10]	33.98 $\pm$ 2.50 / .965	24.39 $\pm$ 2.02 / .827	30.79 $\pm$ 2.13 / .934	9.59	6.40	66.7%
MST-L [10]	34.81 $\pm$ 2.11 / .973	24.23 $\pm$ 1.97 / .808	32.22 $\pm$ 2.04 / .941	10.58	7.99	75.5%

3.5 Evaluation Metrics

We report three standard image quality metrics:

- **PSNR** (peak signal-to-noise ratio, dB): pixel-level fidelity, computed per-channel and averaged.
- **SSIM** (structural similarity index): perceptual structural quality [22].
- **SAM** (spectral angle mapper, degrees): spectral fidelity, reported for CASSI only.

4 Experimental Results

4.1 CASSI Results

Table 1 presents the CASSI benchmark results across 10 KAIST scenes (fig. 1a).

Key findings. The CASSI results reveal a striking dichotomy between mask-aware and mask-oblivious architectures. **MST-L** achieves the best ideal performance (34.81 dB) and the highest oracle recovery (+7.99 dB, $\rho = 75.5\%$), making it the optimal choice when calibration is available. **HDNet**, which processes only the initial spectral estimate without mask input, shows *zero* oracle

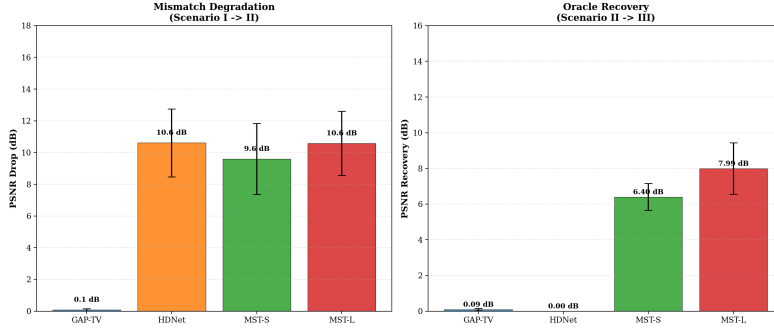


Fig. 2: CASSI mismatch degradation (left, Δ_{deg}) and oracle recovery (right, Δ_{rec}) per method. HDNet shows zero recovery due to its mask-oblivious architecture. MST-L achieves the highest recovery (+7.99 dB), demonstrating the value of mask-aware reconstruction when calibration is available.

Table 2: CACTI reconstruction results (6 videos, $256 \times 256 \times 8$). PSNR (dB) / SSIM reported as mean $\pm$ std. CACTI exhibits the most severe mismatch degradation of all three modalities.

Method	Scenario I	Scenario II	Scenario III	Δ_{deg}	Δ_{rec}	ρ
GAP-TV [7]	26.75 $\pm$ 4.48 / .848	15.81 $\pm$ 1.98 / .305	26.01 $\pm$ 3.72 / .794	10.94	10.21	93.3%
PnP-FFDNet [16]	29.28 $\pm$ 5.53 / .890	11.43 $\pm$ 2.71 / .216	25.39 $\pm$ 3.52 / .820	17.85	13.96	78.2%
ELP-Unfolding [13]	34.09 $\pm$ 4.11 / .965	15.47 $\pm$ 1.71 / .308	29.40 $\pm$ 3.15 / .927	18.63	13.93	74.8%
EfficientSCI [12]	35.39 $\pm$ 4.46 / .973	14.81 $\pm$ 2.19 / .303	27.38 $\pm$ 3.52 / .927	20.58	12.57	61.1%

gain ($\Delta_{\text{rec}} = 0.00$ dB), confirming that mask-oblivious architectures cannot benefit from operator calibration. **GAP-TV** shows negligible mismatch degradation ($\Delta_{\text{deg}} = 0.08$ dB) due to its low overall reconstruction quality (20.37 dB)—the operator mismatch is dominated by the reconstruction noise floor. Under Scenario II (uncorrected mismatch), all deep learning methods converge to a similar performance range (24.05–24.39 dB), erasing the 10 dB advantage they hold under ideal conditions.

SSIM and SAM analysis. SSIM trends mirror PSNR findings. MST-L recovers from 0.808 to 0.941 SSIM under oracle correction, while HDNet remains fixed at 0.813. For SAM, MST-L improves from 15.93° (Scenario II) to 8.22° (Scenario III), approaching the ideal value of 7.44° . This indicates that operator mismatch corrupts spectral fidelity, which mask-aware methods can recover.

4.2 CACTI Results

Table 2 presents the CACTI benchmark results across 6 standard benchmark videos (fig. 1b).

Table 3: SPC reconstruction results (11 Set11 images, 256×256 , 25% sampling). PSNR (dB) / SSIM reported as mean $\pm$ std.

Method	Scenario I	Scenario II	Scenario III	Δ_{deg}	Δ_{rec}	ρ
FISTA-TV [8]	28.06 $\pm$ 3.38 / .852	18.51 $\pm$ 0.69 / .586	26.21 $\pm$ 2.28 / .759	9.55	7.71	80.7%
ISTA-Net [14]	31.85 $\pm$ 3.11 / .916	19.02 $\pm$ 0.61 / .584	27.45 $\pm$ 1.32 / .760	12.83	8.43	65.7%
HATNet [15]	30.98 $\pm$ 0.95 / .847	19.40 $\pm$ 0.59 / .648	29.78 $\pm$ 0.81 / .807	11.58	10.38	89.6%

Key findings. CACTI exhibits the most severe mismatch degradation of all three modalities, with losses ranging from 10.94 dB (GAP-TV) to 20.58 dB (EfficientSCI). The 8-parameter mismatch space—encompassing spatial, temporal, and radiometric errors—creates compounding degradation that is far more destructive than the 3-parameter spatial mismatch in CASSI. Under Scenario II, all methods collapse to 11–16 dB, with SSIM dropping below 0.31, indicating near-complete reconstruction failure.

GAP-TV achieves the highest recovery ratio ($\rho = 93.3\%$), recovering 10.21 dB of its 10.94 dB loss. This demonstrates that classical iterative methods, which directly incorporate the forward operator in each iteration, are highly responsive to operator correction. **EfficientSCI**, despite achieving the best ideal performance (35.39 dB), has the lowest recovery ratio (61.1%), suggesting that its learned features are partially coupled to the ideal operator in ways that oracle mask knowledge cannot fully address.

An inverse relationship. A notable pattern emerges: methods with higher ideal performance suffer larger mismatch degradation and achieve lower recovery ratios. EfficientSCI (best ideal: 35.39 dB) loses 20.58 dB and recovers 61.1%; GAP-TV (worst ideal: 26.75 dB) loses 10.94 dB and recovers 93.3%. This suggests that higher-capacity learned representations encode stronger implicit assumptions about the operator, making them more fragile when those assumptions are violated.

4.3 SPC Results

Table 3 presents the SPC benchmark results across 11 Set11 images (fig. 1c).

Key findings. The SPC results show that gain drift uniformly degrades all methods to a narrow 18.51–19.40 dB range under Scenario II, despite a 3.79 dB spread under ideal conditions. **HATNet** achieves the highest oracle recovery (+10.38 dB, $\rho = 89.6\%$), recovering nearly to its ideal performance (29.78 vs. 30.98 dB). **ISTA-Net**, despite the best ideal performance (31.85 dB), achieves only 65.7% recovery ratio, consistent with the CACTI observation that higher-capacity learned representations are more fragile under mismatch. **FISTA-TV** achieves a balanced 80.7% recovery ratio with the lowest ideal performance.

Table 4: Cross-modality summary. For each modality, we report the range of mismatch degradation (Δ_{deg}), oracle recovery (Δ_{rec}), and the best recovery ratio (ρ_{best}) across methods. Mismatch dim. is the number of mismatch parameters.

Modality	Mismatch dim.	Δ_{deg} range	Δ_{rec} range	ρ_{best}	Best method
CASSI	3	0.08–10.61	0.00–7.99	75.5%	MST-L
CACTI	8	10.94–20.58	10.21–13.96	93.3%	GAP-TV
SPC	1	9.55–12.83	7.71–10.38	89.6%	HATNet

Table 5: Mismatch parameters per modality. CASSI: mask spatial misalignment. CACTI: spatial + temporal + radiometric mismatch (8 parameters). SPC: multiplicative gain drift.

Modality	Parameter	Value	Physical interpretation
CASSI	dx	0.5 px	Horizontal mask shift
	dy	0.3 px	Vertical mask shift
	θ	0.1°	Mask rotation
CACTI	dx	0.5 px	Horizontal mask shift
	dy	0.3 px	Vertical mask shift
	θ	0.1°	Mask rotation
	Δt	0.05	Clock offset
	η	0.95	Duty cycle deviation
	g	1.02	Detector gain
	o	0.002	Detector offset
SPC	σ_n	1.0	Measurement noise
	α	0.0015	Gain drift magnitude
	σ_y	0.03	Measurement noise

Mismatch uniformity. Under Scenario II, the standard deviation across images decreases dramatically (from 0.95–3.38 dB to 0.59–0.69 dB), indicating that gain drift mismatch creates a performance floor independent of image content. This is consistent with the multiplicative nature of gain drift, which affects all measurement rows similarly.

4.4 Cross-Modality Analysis

Table 4 synthesizes the key metrics across all three modalities.

Mismatch severity ranking. CACTI suffers the most severe degradation (10.94–20.58 dB), followed by SPC (9.55–12.83 dB) and CASSI (0.08–10.61 dB). The CACTI severity is driven by its high-dimensional mismatch space (8 parameters vs. 3 for CASSI and effectively 1 for SPC), where spatial, temporal, and radiometric errors compound. For CASSI, the wide range reflects the architectural divide: GAP-TV’s low baseline quality makes it insensitive (0.08 dB), while mask-aware deep networks lose over 10 dB.

Recovery potential. The best recovery ratio varies by modality: GAP-TV achieves 93.3% on CACTI, HATNet achieves 89.6% on SPC, and MST-L achieves 75.5% on CASSI. This suggests that the recovery potential depends on both the method architecture and the mismatch structure. Notably, the best-recovering method is not always the one with the highest ideal performance—a finding with practical implications for system design.

Architectural patterns. Across modalities, we observe consistent patterns: (i) classical methods (GAP-TV, FISTA-TV) show moderate degradation and high recovery ratios; (ii) mask-aware deep methods (MST, ELP-Unfolding, HATNet) show high degradation but substantial recovery; (iii) mask-oblivious deep methods (HDNet) show moderate degradation but zero recovery. This three-way classification provides a practical taxonomy for selecting reconstruction methods based on whether calibration is available.

5 Discussion

Classical vs. deep learning robustness. A consistent finding across all modalities is that classical optimization methods are more robust to operator mismatch than deep learning methods. GAP-TV loses only 0.08 dB on CASSI and 10.94 dB on CACTI (vs. 10.58 dB and 20.58 dB for the best deep networks). However, this robustness comes at the cost of significantly lower ideal performance (20.37 dB vs. 34.81 dB on CASSI). The practical implication is that when calibration is unavailable or imperfect, classical methods may outperform deep learning: on CACTI Scenario II, GAP-TV (15.81 dB) outperforms PnP-FFDNet (11.43 dB) by 4.38 dB despite being 2.53 dB worse under ideal conditions.

The mask-awareness spectrum. Our results reveal that reconstruction methods exist on a spectrum of mask-awareness:

- *Mask-oblivious* (HDNet): The mask is not an input; reconstruction quality depends only on the measurement. No calibration benefit is possible ($\rho = 0\%$).
- *Mask-conditioned* (MST-S, MST-L, HATNet): The mask explicitly conditions the reconstruction network. These methods achieve high calibration gains ($\rho = 67\text{--}90\%$) but suffer the largest mismatch degradation.
- *Operator-iterative* (GAP-TV, FISTA-TV): The forward operator is used in each optimization iteration. These methods achieve moderate-to-high recovery ratios ($\rho = 81\text{--}93\%$) with moderate ideal performance.

This taxonomy has design implications: mask-conditioned architectures are optimal when calibration is feasible, while mask-oblivious architectures provide a stable (but suboptimal) fallback.

Oracle recovery as calibration upper bound. Scenario III provides the upper bound for calibration benefit, achievable only with perfect knowledge of the true operator. In practice, calibration algorithms estimate the mismatch parameters with finite precision, so the actual calibration gain will be less than Δ_{rec} . The large recovery values we observe (up to 13.96 dB for CACTI, 10.38 dB for SPC, 7.99 dB for CASSI) strongly motivate the development of practical calibration methods, even if they recover only a fraction of the oracle bound.

Residual gap analysis. The residual gap $\Delta_{\text{res}} = \text{PSNR}_{\text{I}} - \text{PSNR}_{\text{III}}$ represents unrecoverable losses due to measurement corruption (noise from the mismatched acquisition). For CASSI MST-L, $\Delta_{\text{res}} = 2.59$ dB; for CACTI GAP-TV, $\Delta_{\text{res}} = 0.74$ dB; for SPC HATNet, $\Delta_{\text{res}} = 1.20$ dB. These small residual gaps confirm that operator mismatch—not measurement noise—is the dominant degradation source, validating our low-noise experimental design.

Method-specific insights. **HDNet** presents an instructive case: its mask-oblivious architecture processes only the initial spectral estimate (28 channels from the shift-back operation), making it inherently insensitive to the reconstruction mask. While this provides stability under mismatch (24.05 dB in both Scenarios II and III), it permanently forfeits the 10.61 dB that calibration could recover. **EfficientSCI** on CACTI reveals that state-of-the-art performance under ideal conditions can be a liability: its 35.39 dB ideal performance drops to 14.81 dB under mismatch, a 20.58 dB loss that is only 61.1% recoverable. This suggests that EfficientSCI encodes strong implicit assumptions about the CACTI operator that extend beyond what oracle mask knowledge can correct.

6 Conclusion

We have presented InverseNet, the first cross-modality benchmark for operator mismatch in compressive imaging. By evaluating 11 reconstruction methods across CASSI, CACTI, and SPC under a standardized three-scenario protocol, we establish several key findings: (1) operator mismatch degrades deep learning methods by 10–21 dB, collapsing the performance advantage over classical methods; (2) mask-aware architectures can recover 65–90% of mismatch losses through oracle calibration; (3) mask-oblivious architectures provide mismatch stability at the cost of zero calibration benefit; (4) CACTI’s high-dimensional mismatch space creates the most severe degradation across modalities.

These findings have direct implications for the design of compressive imaging systems. When calibration is feasible, mask-conditioned deep networks (MST-L for CASSI, HATNet for SPC) should be preferred for their high recovery potential. When calibration is unavailable, classical methods (GAP-TV) provide the most robust baseline.

Dataset release. We release all 240+ reconstruction arrays, per-scene metrics, and analysis code. The benchmark is extensible: new modalities, methods, and mismatch models can be added by implementing the three-scenario protocol.

Future directions. InverseNet evaluates oracle (perfect) calibration; a natural extension is to benchmark practical calibration algorithms, measuring how close they come to the oracle bound. Additionally, incorporating learned robustness through mismatch-aware training could bridge the gap between mask-conditioned and mask-oblivious architectures.

References

1. Wagadarikar, A., John, R., Willett, R., Brady, D.: Single disperser design for coded aperture snapshot spectral imaging. *Applied Optics* **47**(10), B44–B51 (2008)
2. Meng, Z., Ma, J., Yuan, X.: End-to-end low cost compressive spectral imaging with spatial-spectral self-attention. In: ECCV. pp. 187–204 (2020)
3. Llull, P., Liao, X., Yuan, X., Yang, J., Kittle, D., Carin, L., Sapiro, G., Brady, D.J.: Coded aperture compressive temporal imaging. *Optics Express* **21**(9), 10526–10545 (2013)
4. Yuan, X., Brady, D.J., Katsaggelos, A.K.: Snapshot compressive imaging: Theory, algorithms, and applications. *IEEE Signal Processing Magazine* **38**(2), 65–88 (2021)
5. Duarte, M.F., Davenport, M.A., Takhar, D., Laska, J.N., Sun, T., Kelly, K.F., Baraniuk, R.G.: Single-pixel imaging via compressive sampling. *IEEE Signal Processing Magazine* **25**(2), 83–91 (2008)
6. Edgar, M.P., Gibson, G.M., Padgett, M.J.: Principles and prospects for single-pixel imaging. *Nature Photonics* **13**(1), 13–20 (2019)
7. Yuan, X.: Generalized alternating projection based total variation minimization for compressive sensing. In: ICIP. pp. 2539–2543 (2016)
8. Beck, A., Teboulle, M.: A fast iterative shrinkage-thresholding algorithm for linear inverse problems. *SIAM Journal on Imaging Sciences* **2**(1), 183–202 (2009)
9. Boyd, S., Parikh, N., Chu, E., Peleato, B., Eckstein, J.: Distributed optimization and statistical learning via the alternating direction method of multipliers. *Foundations and Trends in Machine Learning* **3**(1), 1–122 (2011)
10. Cai, Y., Lin, J., Hu, X., Wang, H., Yuan, X., Zhang, Y., Timofte, R., Van Gool, L.: Mask-guided spectral-wise transformer for efficient hyperspectral image reconstruction. In: CVPR. pp. 17502–17511 (2022)
11. Hu, X., Cai, Y., Lin, J., Wang, H., Yuan, X., Zhang, Y., Timofte, R., Van Gool, L.: HDNet: High-resolution dual-domain learning for spectral compressive imaging. In: CVPR. pp. 17542–17551 (2022)
12. Wang, L., Cao, M., Yuan, X.: EfficientSCI: Densely connected network with space-time factorization for large-scale video snapshot compressive imaging. In: CVPR. pp. 18477–18486 (2023)
13. Yang, C., Zhang, S., Yuan, X.: ELP-Unfolding: Efficient long-range perception unfolding network for compressive video sensing. In: ECCV. pp. 238–254 (2023)
14. Zhang, J., Ghanem, B.: ISTA-Net: Interpretable optimization-inspired deep network for image compressive sensing. In: CVPR. pp. 1828–1837 (2018)
15. Song, J., Chen, B., Zhang, J.: Memory-augmented deep unfolding network for compressive sensing. In: ACM MM. pp. 4249–4258 (2023)
16. Yuan, X., Liu, Y., Suo, J., Dai, Q.: Plug-and-play algorithms for large-scale snapshot compressive imaging. In: CVPR. pp. 1447–1457 (2020)
17. Choi, I., Kim, M.H., Gutierrez, D., Jeon, D.S., Nam, G.: High-quality hyperspectral reconstruction using a spectral prior. *ACM Transactions on Graphics* **36**(6), 218:1–218:13 (2017)

18. Kulkarni, K., Lohit, S., Turaga, P., Kerviche, R., Ashok, A.: ReconNet: Non-iterative reconstruction of images from compressively sensed measurements. In: CVPR. pp. 449–458 (2016)
19. Zbontar, J., Knoll, F., Sriram, A., Murrell, T., Huang, Z., Muckley, M.J., Defazio, A., Stern, R., Johnson, P., Bruno, M., et al.: fastMRI: An open dataset and benchmarks for accelerated MRI. arXiv preprint arXiv:1811.08839 (2018)
20. Elser, V.: Phase retrieval by iterated projections. *Journal of the Optical Society of America A* **20**(1), 40–55 (2003)
21. Arguello, H., Rueda, H., Wu, Y., Prather, D.W., Arce, G.R.: Higher-order computational model for coded aperture spectral imaging. *Applied Optics* **52**(10), D12–D21 (2013)
22. Wang, Z., Bovik, A.C., Sheikh, H.R., Simoncelli, E.P.: Image quality assessment: from error visibility to structural similarity. *IEEE Transactions on Image Processing* **13**(4), 600–612 (2004)